

Strict Cone Separation from Positive KKT Certificates: a Gordan–Farkas Mechanism and a Local Sharpness Theorem

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Abstract

We isolate a reusable mechanism that converts a *strictly positive* KKT certificate into a strictly quantitative separation statement, and we determine its sharp constant in a concrete non-smooth extremal problem. Let $G \in \mathbb{R}^{m \times n}$ admit $y > 0$ with $G^\top y = 0$. Gordan’s theorem of alternatives then forces the descent cone to collapse, $\{h : Gh \leq 0\} = \{0\}$, hence $\text{rank } G = n$, and the column space contains no nonzero nonnegative vector; consequently the augmented matrix $[G \mid -e]$ with $e \geq 0$, $e \neq 0$ is nonsingular. For the associated linear program $\min c$ subject to $v_i h \leq -1$ and $w_q h \leq c$ we prove $c_* = 1/L$ with $L = \sum_q \lambda_q$, where λ are the positive multipliers of $G^\top y = 0$; the dual-optimal Farkas certificate is exactly $c_*(\omega, \lambda)$. Quantitatively, the sharpness constant $\eta = \min_{\|u\|=1} \max_i (Gu)_i$ is certified strictly positive ($\eta \geq 0.0766443030678\dots$ in the instance), and a second-order Taylor bound with explicit constant $K = 648$ yields a certified local radius $\rho_{\text{cert}} = \eta/K = 1.18278 \times 10^{-4}$ inside which the local maximum strictly exceeds its boundary value. Finally we characterise the *scope boundary* of the mechanism: the certificate is a positive combination and therefore belongs to the sum-of-nonnegative type; canonical involution-invariant data cannot yield the one-sided, axis-breaking separator that an arithmetic realisation would require, and the orientation available at the membership boundary we examine is Archimedean in origin. Thus the mechanism is a strictly proved *local geometric* separation statement, and it does not by itself upgrade to an arithmetic or spectral one. No claim about the Riemann Hypothesis is made or used.

1 Setting and hypotheses

Fix $n = 5$, $m = n + 1 = 6$, and a sign vector $\sigma = (\sigma_1, \dots, \sigma_5) \in \{\pm 1\}^5$. For $\phi = (\phi_1, \dots, \phi_5) \in (0, \pi/2)^5$ put

$$F_k(\phi) = \sum_{j=1}^5 \gamma_j \cos(k\phi_j), \quad \gamma_j = \begin{cases} \sigma_j, & k \text{ odd}, \\ 1, & k \text{ even}, \end{cases} \quad (1)$$

so that F_k is a trigonometric polynomial of degree k . Let ϕ^* be a point at which the *six-active* conditions hold for the odd frequencies 13, 19 and the even functionals with $2q \in \{6, 8, 14, 18\}$, i.e. frequencies 12, 16, 28, 36. Define the signed active functions

$$g_{13} = -F_{13}, \quad g_{19} = F_{19}, \quad g_q = F_q - \frac{1}{2} \quad (q \in \mathcal{A} := \{6, 8, 14, 18\}). \quad (2)$$

Write $G \in \mathbb{R}^{6 \times 5}$ for the matrix whose rows are the gradients $\nabla g_i(\phi^*)$, ordered as (13, 19, 6, 8, 14, 18).

(H1) (*strict positive KKT certificate.*) There is $y \in \mathbb{R}^6$ with $y > 0$ componentwise and $G^\top y = 0$; we write $y = (\omega_{13}, \omega_{19}, \lambda_6, \lambda_8, \lambda_{14}, \lambda_{18})$ and $L = \sum_{q \in \mathcal{A}} \lambda_q > 0$.

(H2) (*normalisation.*) $\omega_{13} + \omega_{19} = 1$.

(H3) (*attainment of the active square system.*) There exist $h^* \in \mathbb{R}^5$ and $c^* \in \mathbb{R}$ with $v_i h^* = -1$ for $i \in \{13, 19\}$ and $w_q h^* = c^*$ for all $q \in \mathcal{A}$, where v_i, w_q are the corresponding rows of G .

In the concrete instance we use, $\sigma = (-1, 1, 1, -1, 1)$ and the numerical realisation of ϕ^* gives (to the stated precision)

$$L = 1.99676941227516035204325246457, \quad c^* = 0.500808953629041885590052265792.$$

Hypothesis (H1) is a strict-complementarity/positive-KKT hypothesis: it holds at the point considered and is *not* automatic at degenerate points (see §5).

2 The abstract mechanism

We use the following classical alternative; see e.g. Gordan's original note and standard texts.

Theorem 2.1 (Gordan, [1, 2]). *For $G \in \mathbb{R}^{m \times n}$, exactly one of the following holds: (i) there is $h \neq 0$ with $Gh \leq 0$; (ii) there is $y > 0$ with $G^\top y = 0$.*

Lemma 2.2 (Trivial descent cone). *Assume (H1). Then $\{h \in \mathbb{R}^n : Gh \leq 0\} = \{0\}$.*

Proof. By (H1), alternative (ii) of Theorem 2.1 holds; hence (i) fails, which is the assertion. $\square$

Corollary 2.3 (Structural full column rank). *Assume (H1). Then $\text{rank } G = n$.*

Proof. If $\text{rank } G < n$ then $\ker G \neq \{0\}$; choosing $0 \neq h \in \ker G$ gives $Gh = 0 \leq 0$, contradicting Lemma 2.2. $\square$

Corollary 2.4 (No nonzero nonnegative vector in the column space). *Assume (H1). Then $\text{col}(G) \cap \{z \geq 0, z \neq 0\} = \emptyset$. In particular, for every $e \geq 0, e \neq 0$, one has $e \notin \text{col}(G)$.*

Proof. Suppose $0 \neq z \geq 0$ and $z \in \text{col}(G)$. Since $\text{col}(G)$ is a linear subspace, $-z \in \text{col}(G)$ as well, so there is h with $Gh = -z \leq 0$. Moreover $h \neq 0$, because $z \neq 0$. This contradicts Lemma 2.2. $\square$

Note that Corollary 2.4 yields nonsingularity of the augmented matrix *structurally*, i.e. from the positive certificate and not from a numerical determinant:

$$m = n + 1, \quad \text{rank } G = n, \quad e \notin \text{col}(G) \implies \text{rank}[G \mid -e] = n + 1. \quad (3)$$

3 The exact bridge constant

Consider the linear program

$$c_* = \min_{h, c} c \quad \text{subject to} \quad v_i h \leq -1 \quad (i \in \{13, 19\}), \quad w_q h - c \leq 0 \quad (q \in \mathcal{A}). \quad (4)$$

Theorem 3.1 (Exact bridge constant). *Assume (H1), (H2) and (H3). Then*

$$c_* = \frac{1}{L}, \quad L = \sum_{q \in \mathcal{A}} \lambda_q.$$

Moreover the Farkas dual certificate $(\nu, \mu) = (\omega/L, \lambda/L)$ is dual-optimal and satisfies $(\nu, \mu) = c_(\omega, \lambda)$.*

Proof. Lower bound. Let h be primal-feasible. Using $v_{13}h \leq -1$, $v_{19}h \leq -1$ and (H2) with $\omega \geq 0$,

$$\omega_{13}(v_{13}h) + \omega_{19}(v_{19}h) \leq -(\omega_{13} + \omega_{19}) = -1.$$

By (H1), $\omega_{13}v_{13} + \omega_{19}v_{19} = -\sum_q \lambda_q w_q$, hence $\sum_q \lambda_q w_q h \geq 1$. Since $w_q h \leq \max_r w_r h$ and $\lambda \geq 0$, $L \max_q (w_q h) \geq \sum_q \lambda_q w_q h \geq 1$, so $c_* \geq 1/L$.

Upper bound. Let (h^*, c^*) be as in (H3) and take the inner product of $G^\top y = 0$ with h^* . The odd components contribute $-\omega_{13} - \omega_{19} = -1$ and the even components $\sum_q \lambda_q w_q h^* = c^* L$, so $-1 + c^* L = 0$, i.e. $c^* = 1/L$. By (H3), h^* is primal-feasible, whence $c_* \leq c^* = 1/L$.

Combining the two bounds gives $c_* = 1/L$. Finally, $(\nu, \mu) = (\omega/L, \lambda/L)$ satisfies $\nu, \mu \geq 0$, $\sum_q \mu_q = 1$ and $\nu_{13}v_{13} + \nu_{19}v_{19} + \sum_q \mu_q w_q = 0$, so it is dual-feasible with value $\nu_{13} + \nu_{19} = 1/L = c_*$; dual optimality follows, and the displayed identity $(\nu, \mu) = c_*(\omega, \lambda)$ is immediate. $\square$

Remark 3.2. Theorem 3.1 shows that c_* is not an isolated numerical constant: it is the reciprocal of the sum of the active even multipliers, and the Farkas separator is the KKT certificate rescaled by c_* .

4 Quantitative sharpness and a certified local radius

Define the sharpness constant

$$\eta = \min_{\|u\|_2=1} \max_i (Gu)_i. \quad (5)$$

By Lemma 2.2, $\max_i (Gu)_i > 0$ for every $u \neq 0$, and the unit sphere is compact, so $\eta > 0$; this is a second, quantitative consequence of the positive certificate.

To certify a numerical lower bound we exploit the sign decomposition: every $u \neq 0$ is uniquely $u = s \odot w$ with $s \in \{\pm 1\}^5$, $w \geq 0$, $\sum_j w_j = 1$, and $\max_i (Gu)_i / \|u\|_2 \geq \max_i (Gu)_i / \|u\|_1$. Hence $\eta_1 = \min_s \min\{\max_i (G(s \odot w))_i : w \geq 0, \mathbf{1}^\top w = 1\}$ satisfies $\eta \geq \eta_1$, and each inner problem is a linear program. By weak duality, for any y in the simplex, $\max_i (G(s \odot w))_i \geq \min_j ((G_s)^\top y)_j$, which requires only $y \geq 0$ and $\sum_i y_i = 1$. Enclosing the coefficients rationally gives the certified bound

$$\boxed{\eta \geq \underline{\eta} = 0.076644303067798235521 \dots} > 0. \quad (6)$$

For the Taylor step, work in the coordinates ϕ . Since $\partial^2 F_k / \partial \phi_j^2 = -k\gamma_j \cos(k\phi_j)$, the Hessian is diagonal with $|\partial^2 F_k / \partial \phi_j^2| \leq k^2$, hence for the even rows $k \leq 36$ and the odd rows $k \leq 23$, the remainder satisfies $|R_i(h)| \leq K\|h\|_2^2$ with

$$K = \frac{1}{2} \cdot 36^2 = 648. \quad (7)$$

Writing $g_i(\phi^* + h) = g_i(\phi^*) + \nabla g_i(\phi^*) \cdot h + R_i(h)$ and using $g_{13}(\phi^*) = g_{19}(\phi^*) = c_0$, $g_q(\phi^*) = 0$, we obtain for every $h \neq 0$

$$\max_i g_i(\phi^* + h) \geq c_0 + \eta\|h\|_2 - K\|h\|_2^2. \quad (8)$$

Consequently

$$\boxed{0 < \|h\|_2 < \rho_{\text{cert}} := \frac{\eta}{K} \implies \max_i g_i(\phi^* + h) > c_0} \quad (9)$$

and in the instance at hand $\rho_{\text{cert}} = 1.18278 \times 10^{-4}$ (the tightest coordinate half-width in the original variable is 7.99×10^{-5}).

Remark 4.1. Statement (9) is a *local* sharpness theorem: it certifies that ϕ^* is a strict local minimiser of the local maximum, inside a fully explicit neighbourhood. It does not assert global optimality; in particular no equality $V = c_0$ is claimed for the global problem.

5 Scope of the mechanism: an interface obstruction

The mechanism above is a statement of linear-algebraic/convex duality. We record honestly why it does not automatically upgrade to an arithmetic or spectral statement.

Proposition 5.1 (Class of the certificate). *The certificate in (H1) is a positive combination, $\sum_i y_i g_i = 0$ with $y > 0$. Consequently the mechanism belongs to the sum-of-nonnegatives positivity type, and renaming it (e.g. as a “Farkas separation” or “positive KKT certificate”) does not make it a new positivity mechanism.*

Proof. Immediate from the displayed identity, which is exactly (H1) rewritten row-wise. $\square$

Proposition 5.2 (Symmetry obstruction). *Suppose the available arithmetic data are invariant under the involution ι associated with the functional equation (so that condition sets are ι -invariant). Then no one-sided, axis-breaking separator can be produced from them: a symmetric datum cannot generate an asymmetric (one-sided) separation object.*

Proof. If a predicate P satisfies $P(\beta) = P(\iota\beta)$ for all admissible data, then it cannot separate β from $\iota\beta$; a one-sided separator by definition takes different values on the two sides, so it cannot be realised by such P . $\square$

Proposition 5.3 (Archimedean orientation at the membership boundary). *For the membership boundary we examined, the available orientation is Archimedean in origin: the comparison involves an Archimedean constant and an Archimedean iterated logarithm, so the resulting direction is not arithmetic.*

Proof. Direct inspection of the comparison defining the boundary: the two sides involve $\sigma(n)$ and $e^\gamma n \log \log n$ respectively, and e^γ , $\log \log n$ are Archimedean objects. $\square$

Remark 5.4 (Scope boundary). Together, Propositions 5.1–5.3 delimit the scope of the mechanism: it is a strictly proved *local geometric* separation statement with an exact constant, and the passage to arithmetic/spectral content requires an interface that is not supplied here. We state the resulting separation as a scope boundary,

$$\text{local arithmetic structure} \not\approx \text{global spectral localisation},$$

rather than as a claim about any particular spectral problem. No Riemann Hypothesis statement is used or asserted anywhere in this paper.

6 Data and reproducibility

All numerical statements are accompanied by scripts and machine-checked outputs in the accompanying repository: the reconstruction of ϕ^* at high precision, the six-active gradient matrix G , the Gordan certificate y , the 32-sign linear programs certifying (6), and the Taylor/remainder computation. The certified lower bound η is obtained from dual feasible points (only $y \geq 0$, $\sum_i y_i = 1$ are used) together with a rational enclosure of the coefficients; the margin used in the enclosure is 10^{-30} , against a truncation error of order 10^{-45} at the working precision.

References

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